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International Mathematics Assessments for Schools

2020 ~ 2021 JUNIOR DIVISION FIRST ROUND PAPER

Time allowed：75 minutes

When your teacher gives the signal, begin working on the problems.

INSTRUCTION AND INFORMATION

GENERAL

1. Do not open the booklet until told to do so by your teacher.
2. No calculators, slide rules, log tables, math stencils, mobile phones or other calculating aids are permitted. Scribbling paper, graph paper, ruler and compasses are permitted, but are not essential.
3. Diagrams are NOT drawn to scale. They are intended only as aids.
4. There are 20 multiple-choice questions, each with 5 choices. Choose the most reasonable answer. The last 5 questions require whole number answers between 000 and 999 inclusive. The questions generally get harder as you work through the paper. There is no penalty for an incorrect response.
5. This is a mathematics assessment, not a test; do not expect to answer all questions.
6. Read the instructions on the answer sheet carefully. Ensure your name, school name and school year are filled in. It is your responsibility that the Answer Sheet is correctly coded.

THE ANSWER SHEET

1. Use only pencils.
2. Record your answers on the reverse side of the Answer Sheet (not on the question paper) by FULLY filling in the circles which correspond to your choices.
3. Your Answer Sheet will be read by a machine. The machine will see all markings even if they are in the wrong places. So please be careful not to doodle or write anything extra on the Answer Sheet. If you want to change an answer or remove any marks, use a plastic eraser and be sure to remove all marks and smudges.

INTEGRITY OF THE COMPETITION

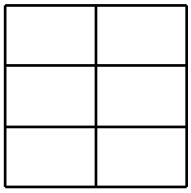
The IMAS reserves the right to re-examine students before deciding whether to grant official status to their scores.

2020 ~ 2021 JUNIOR DIVISION FIRST ROUND PAPER

Questions 1-10, 3 marks each

1. Given that $|x-2019|+(y+2020)^2=0$, what is the numerical value of $(x+y)^y$?
   （A） 1　　（B） –2019　　（C） 0　　（D） –1　　（E） 2020

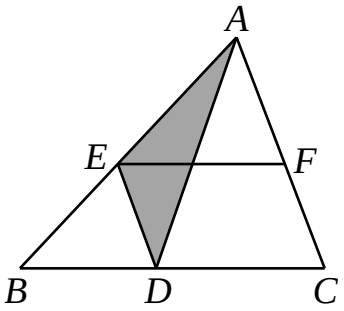
2. A square, with side length 6 cm, is divided into 6 congruent small rectangles as shown in the figure. What is the length of the diagonal, in cm, of each small rectangle?



   （A） 3　　（B） $\sqrt{13}$　　（C） 5　　（D） $2\sqrt{10}$　　（E） $6\sqrt{2}$

3. It is known that the sum of a positive integer and 4 is a multiple of 3. If the sum of the same positive integer and 3 is a multiple of 7, what is the smallest possible value of this positive integer?
   （A） 4　　（B） 7　　（C） 8　　（D） 11　　（E） 32

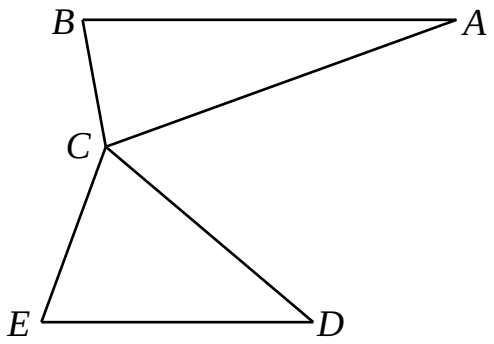
4. In the figure, the area of triangle *ABC* is 2020 cm^2. If points *D*, *E* and *F* are on sides *BC*, *BA* and *AC*, respectively, such that *CDEF* is a parallelogram and its area is $\frac{50}{101}$ of triangle *ABC*. What is the area, in cm^2, of the shaded region?



   （A） 100　　（B） 500　　（C） 505　　（D） 1000　　（E） 1010

5. There are three boxes that are painted red, yellow and blue respectively. We are to put 2 identical apples and 2 identical pears into these three boxes such that no two identical fruit can be placed into the same box. If empty boxes are allowed, how many different placement methods can we have?
   （A） 27　　（B） 18　　（C） 12　　（D） 9　　（E） 6

6. In the figure below, it is given that $AB//DE$, $AB=AC$, $CD=DE$, $\angle B=80°$ and $\angle E=70°$. What is the measure, in degrees, of $\angle ACD$?



（A）50　　（B）55　　（C）60
（D）70　　（E）cannot be determined

---

7. Let x and y be real numbers such that $|x|=5$, $|y|=7$ and $x<y$. What is the value of the algebraic expression x^2+xy-y^2?

（A）-59　　（B）59
（C）11 or 59　　（D）-11 or 59　　（E）11 or -59

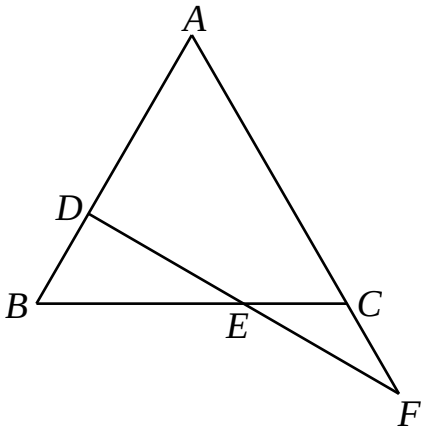
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8. Define the operation $a*b=\dfrac{a-2020}{b+2020}$. If $(a*2020)+(2020*a)=1$, what is the value of a?

（A）8080　　（B）6060　　（C）4040　　（D）2020　　（E）1010

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9. In the figure below, ABC is an equilateral triangle with side length 1 cm, where a straight line intersects AB, BC and the extension of AC at points D, E and F, respectively. If $BD=CE=CF$, what is the length, in cm, of BD?



（A）$\dfrac{1}{3}$　　（B）$\dfrac{1}{4}$　　（C）$\dfrac{1}{2}$　　（D）$\dfrac{\sqrt{3}}{3}$　　（E）$\dfrac{\sqrt{3}}{2}$

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10. If the cost to buy 2 pens, 3 ballpens and 5 notebooks in a stationery store is $44 and the cost to buy 3 pens, 7 ballpens and 10 notebooks is $81. How much does it cost to buy 1 pen, 3 ballpens and 4 notebooks?

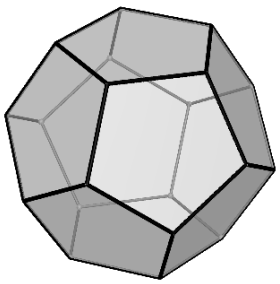
（A）$12　　（B）$25　　（C）$31
（D）$35　　（E）cannot be determined

Questions 11-20, 4 marks each

11. Given two positive integers such that $\frac{4}{7}$ of one number is equivalent to $\frac{2}{5}$ of the other. What is the minimum possible sum of these two numbers?

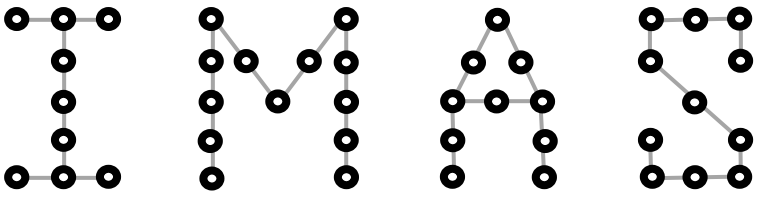
（A）13　　（B）14　　（C）15　　（D）16　　（E）17

12. Andy used some number of regular pentagonal cardboards of the same size to form a solid which is shown on the diagram below. How many edges does this solid have?



（A）10　　（B）15　　（C）20　　（D）25　　（E）30

13. There are 43 pearls with a hole in the center that are to be stringed together using a thread such that the thread will form the letters "I", "M", "A", and "S", as shown in the figure below. If each pearl can only be passed through once by the thread, how many from the four letters cannot be done?



（A）0　　（B）1　　（C）2　　（D）3　　（E）4

14. John wants to know the weight of a bottle of fruit juice using a two-pan balance having the following items, namely a Teddy Bear doll, a Snoopy doll, a Mickey Mouse doll and 5 kg weight as follows:
    - If John places the Teddy Bear doll and the bottle of juice on the left plate, and places both the Snoopy and Mickey Mouse dolls on the right plate, it will be balanced.
    - If John places the Snoopy doll, Mickey Mouse doll and the bottle of juice on the left plate, and places the Teddy Bear doll and the 5 kg weight on the right plate, it will be balanced.

    What is the weight, in kg, of the bottle of juice?

    （A）2.5　（B）3　（C）3.5　（D）4　（E）5

15. Given that a, b and c are positive integers, which among the following values may the expression $(a+b+c)(a+b-c)(a-b+c)(-a+b+c)$ take?

    （A）44　（B）46　（C）48　（D）50　（E）52

16. Using the numbers 1, 2, …, 2020, let the sum of all the numbers that are divisible by 3 be X and the sum of all the numbers that are divisible by 2 but not by 3 be Y. What is the value of $X-Y$?

    （A）1008　（B）−1008　（C）1344　（D）1011　（E）−1011

17. What is the product of all real numbers x that satisfies the expression $(|x-2|-|x-6|)(|x-6|-|x-12|)(|x-12|-|x-2|)=0$?

    （A）28　（B）126　（C）144　（D）252　（E）2016

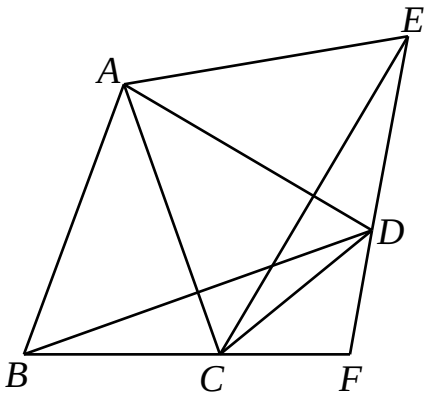
18. How many pairs of positive integers (a, b), where $a<b$, are there which satisfies the expression $\frac{1}{a}+\frac{1}{b}=0.3$?

    （A）1　（B）2　（C）3　（D）4　（E）infinity

19. In the figure on the right, $ABCDE$ is a pentagon where $\angle BAC=\angle CAD=\angle DAE=40°$, $AC\perp BD$ and $AD\perp CE$. Extend sides BC and ED to meet at point F. What is the measure, in degrees, of $\angle BFE$?

    （A）100　（B）90　（C）80

    （D）60　（E）cannot be determined



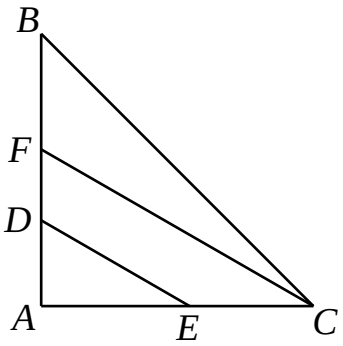
20. Given a four-digit integer $\overline{abcd}$ and a positive integer k which satisfies $\overline{abcd} = k(a+b+c+d)^2$, what is the minimum value of k?

（A）32　　（B）13　　（C）12　　（D）4　　（E）3

Questions 21-25, 6 marks each

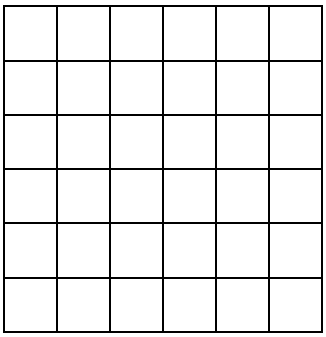
21. Given a positive integer n such that $2020n$ can be expressed as a product of three consecutive positive integers, what is the smallest possible value of n?

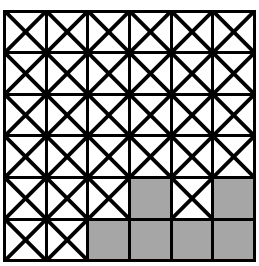
22. In the figure below, ABC is an isosceles triangle where $\angle A$ is a right angle, points D and E are on sides AB and AC respectively such that $AE > AD$. Let F be a point on AB such that CF is parallel to DE. If $\angle ADE = 4\angle BCF$, what is the measure, in degrees, of $\angle AED$?

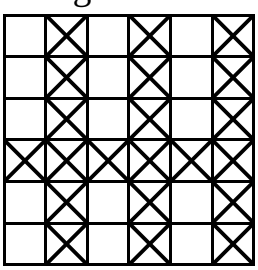


23. Let p, q and r be prime numbers such that $(p^2+q)(q^2+r)(r^2+p) = 32164$. What is the numerical value of the algebraic expression $p^2q + q^2r + r^2p$?

24. We need to remove several unit squares from a 6×6 grid such that there does not exist any area that have 6 unit squares which are connected via common sides. What is the minimum number of unit squares that we must remove? For example, in the lower left figure below, the shaded area is an example of an area that have 6 unit squares which are connected together; while the lower right figure shows an example of a grid which does not have any 6 unit squares that are connected together left.







25. Let the number of odd digits of a positive integer *n* be denoted as $O_{(n)}$ and the number of even digits be denoted as $E_{(n)}$. For example, $O_{(13)} = 2$ and $E_{(58)} = 1$. What is the numerical value of $O_{(1)} - E_{(2)} + O_{(3)} - E_{(4)} + ... + O_{(201)} - E_{(202)}$?
(Note: 0 is even digit)

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